OpenPallet: the standard's cards

OpenPallet is an open exchange format for palletizing data: one JSON file describes the pallet, the product, the slip sheets, the layers and every robot move.

An OpenPallet 1.2 file carries the mark `"format": {"name": "OpenPallet", "version": "1.2"}`, every position in one pallet frame, and every orientation and every direction in degrees counter-clockwise from the top of the drawing.

The same values have a fixed layout for the controller and the robot (PLC structure, CSV), so the PLC and robot program reads what the drawing shows.

1	Station layout and directions
2	Orientation
3	Gripper
4	Open side and alignment
5	Placing direction
6	Stack and slip sheets
7	OpenPallet 1.2 file

Example on the cards: a pallet of Product A (EUR2 1200 × 1000, product 290 × 250 × 200, 6 layers, 48 moves).

OpenPallet, created and maintained by the makers of PalletDesigner.

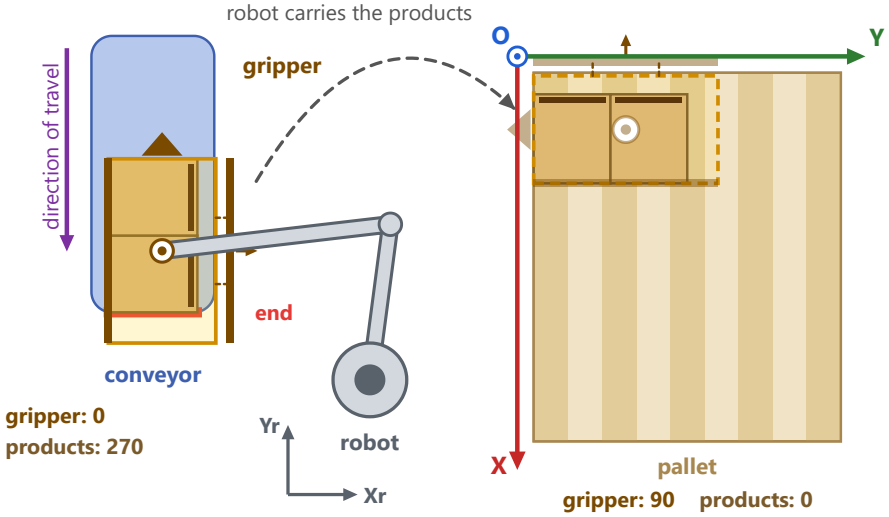
Text and drawings of the cards: licence CC BY 4.0. Every copy, translation or quotation names the author with the sentence above.

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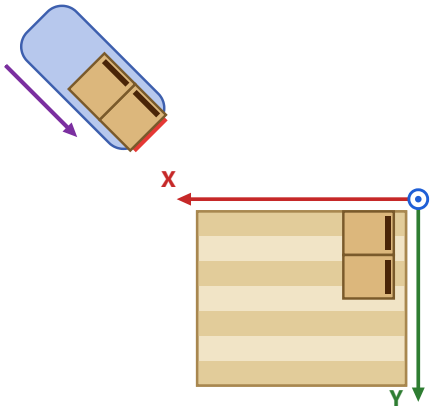
1 Station layout and directions

Every element of the station has its own frame and normal pose. The file gives orientations and positions relative to the normal pose: of the conveyor at the pick, of the pallet at the place, however they stand in the station. Pallet frame: X down the drawing along the length (`pallet.length`), Y to the right along the width (`pallet.width`), Z up.

Station from above
every element in its normal pose

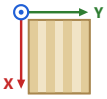
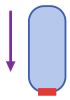


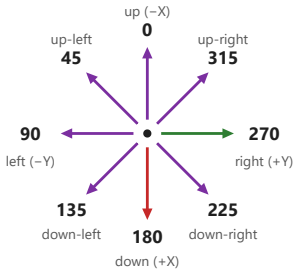
In the station
elements stand any way



the file counts each element
as in its normal pose,
as if the camera turned with it

Example: Product A, unique layer 1, move 1.

Symbol	Element
	**Robot.** The integrator sets its frame (Xr, Yr). The robot program places the pallet and the conveyor in it.
	**Pallet.** Origin O in a corner, X along the length (also when the pallet is wider than long), Y along the width, Z up from the deck boards.
	**Conveyor.** Travel down the drawing, the conveyor end (red line) at the bottom.
	**Gripper** at orientation 0. Triangle: the front. Dark bars: sides 90 and 270. Circle: the robot's flange.
	**Product** at orientation 0. Dark bar: face 0 (the front). Blue bar: the label.



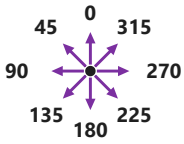
Compass: directions every 45 degrees, numbers increase counter-clockwise.

Direction	On the drawing	Vector (X, Y)
0	up	(−1, 0)
45	up-left	(−1, −1)
90	left	(0, −1)
135	down-left	(+1, −1)
180	down	(+1, 0)
225	down-right	(+1, +1)
270	right	(0, +1)
315	up-right	(−1, +1)
900	straight down	(0, 0)

- **Orientation** (0, 90, 180, 270): the side the front of an element faces. 0 is the normal pose: the product stands as it reaches the conveyor end without being turned, the gripper as it picks most often.
- The file gives only orientations. How to turn the gripper to reach them safely is the robot program's decision.
- **Direction** (compass): the placing direction of the products, in the pallet frame. **900** = neutral: straight down (card 5), no open side and centre (card 4). **999** = not assigned.
- **Move** (`moves[]`): the products the gripper picks and places at once. Z counts from the pallet's deck boards (card 6).

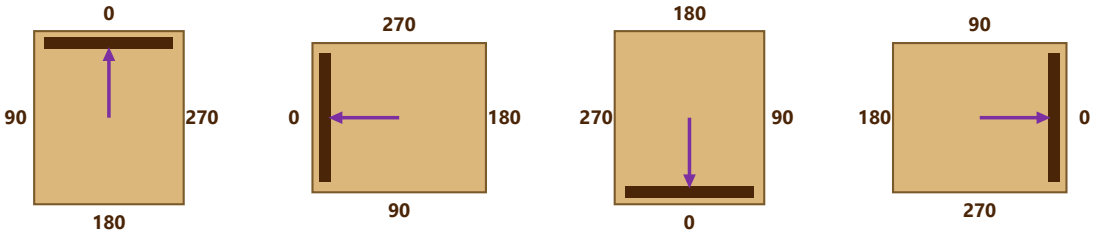
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2 Orientation



Orientation is the side the front of an element (face 0 of a product, the front of the gripper) faces in the frame it lies in: 0, 90, 180 or 270, as on the compass. Orientation 0 is the normal pose.

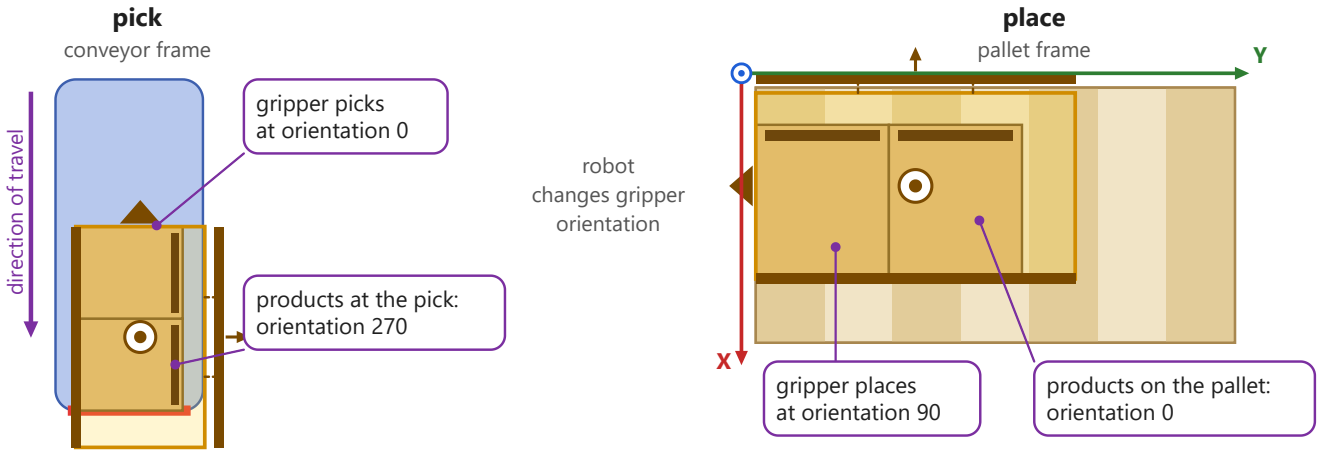
A product in the normal pose (orientation 0) stands as it reaches the conveyor end without being turned: its length along X, face 0 at the top of the drawing.



Numbers beside the faces: the face numbers. Dark bar: face 0.

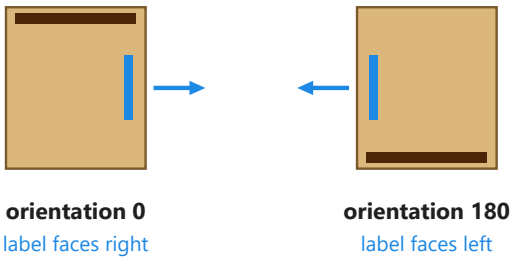
products at the pick → gripper picks → gripper places → products on the pallet

Step	Field in the file	Frame	example 1	example 2
products at the pick	`products_at_pick[k].orientation`	conveyor	270	90
gripper picks	`gripper_pick_orientation`	conveyor	0	0
gripper places	`gripper_place_orientation`	pallet	90	90
products on the pallet	`products[].orientation`	pallet	0	180



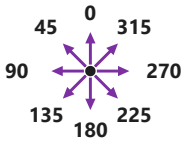
The drawing shows example 1: Product A, unique layer 1, move 1. Example 2 is move 4 of unique layer 2.

- The product turns together with the gripper. The file gives all four orientations of a move: nothing has to be computed.
- The controller (PLC) gets the orientations at the pick and both orientations of the gripper. It does not need the orientation on the pallet: the products lie as the gripper placed them.
- The faces with a label (`label_sides` of the product type) are given in the normal pose, Product A: `[270]`.



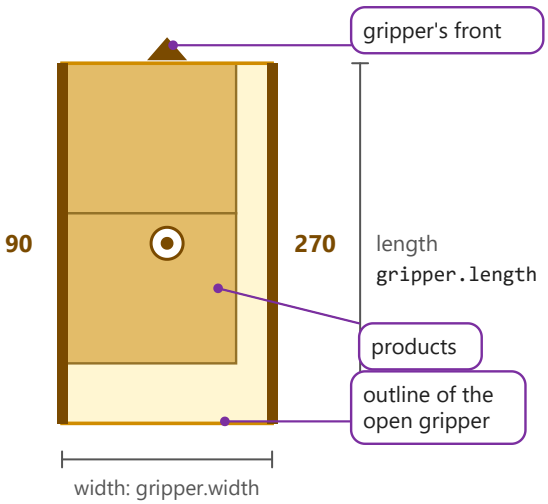
The label of Product A (face 270) turns together with the product.

3 Gripper

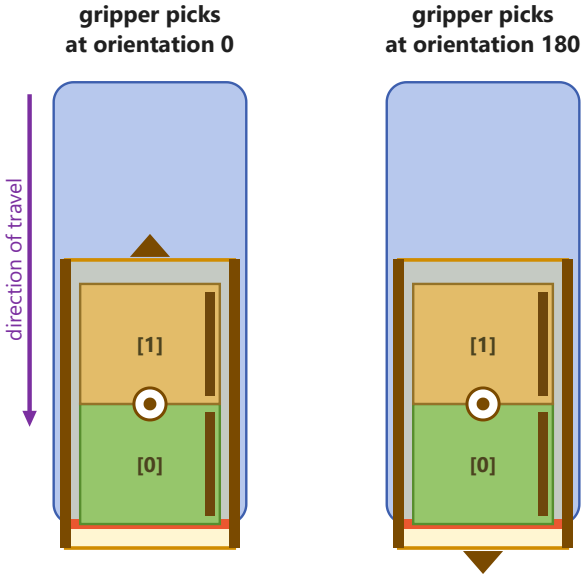


The gripper at orientation 0 stands with its length along X and its front (the triangle) towards the top of the drawing. At the pick its orientation counts in the conveyor's frame, at the place in the pallet's frame.

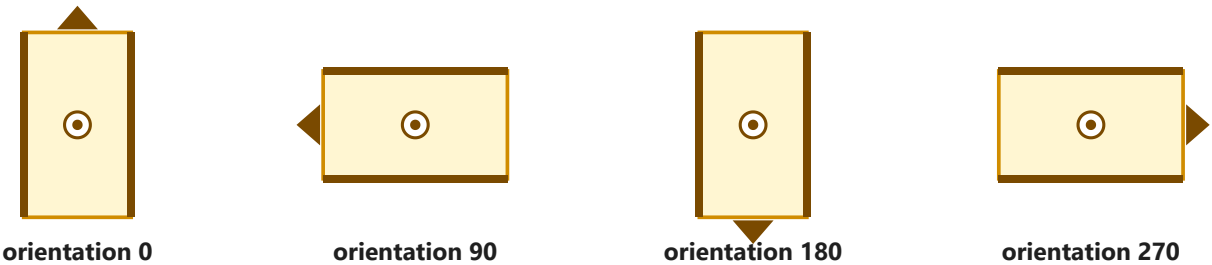
We recommend taking the gripper's orientation 0 as the gripper most often picks the products from the conveyor. The file gives the outline of the open gripper, whatever its kind: for a jaw gripper, the outline with the jaws fully open. The outline is usually larger than the products picked. The open side (90 or 270, card 4) must be free when placing: on a jaw gripper it is the side of the moving jaw, on a fork gripper the side the forks slide in from.



Gripper at orientation 0. Outline of the open gripper in mm: length along, width across.



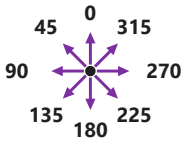
Conveyor: travel down the drawing, its end at the bottom. [0]: the product at the conveyor end.



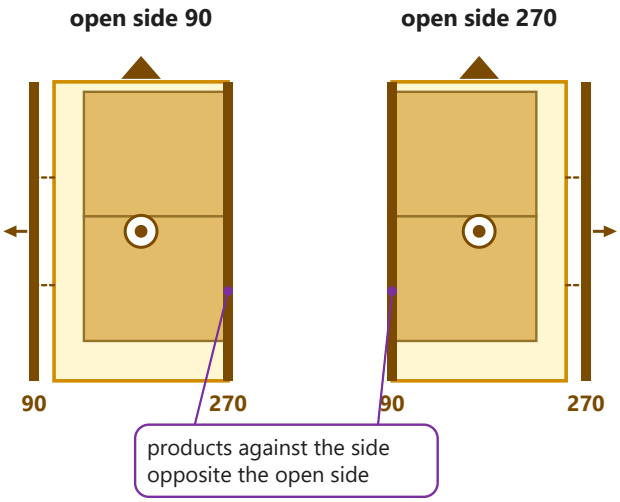
At the place the gripper can stand at any of the four orientations in the pallet frame.

- Gripper orientation at the pick (`gripper_pick_orientation`): 0 or 180, in the conveyor's frame.
- Gripper orientation at the place (`gripper_place_orientation`): 0, 90, 180 or 270, in the pallet frame.
- Products at the pick (`products_at_pick`): the products in the order they reach the conveyor end, each with its product number (`product_id`) and its orientation at the pick (`orientation`). `[0]` is the product at the conveyor end, then the next ones from the end.
- At orientation 0 the back of the gripper lies at the conveyor end, at orientation 180 its front.
- The allowed gripper orientations at the pick (`gripper.allowed_pick_orientation`) and at the place (`gripper.allowed_place_orientation`) are settings of the station.

4 Open side and alignment

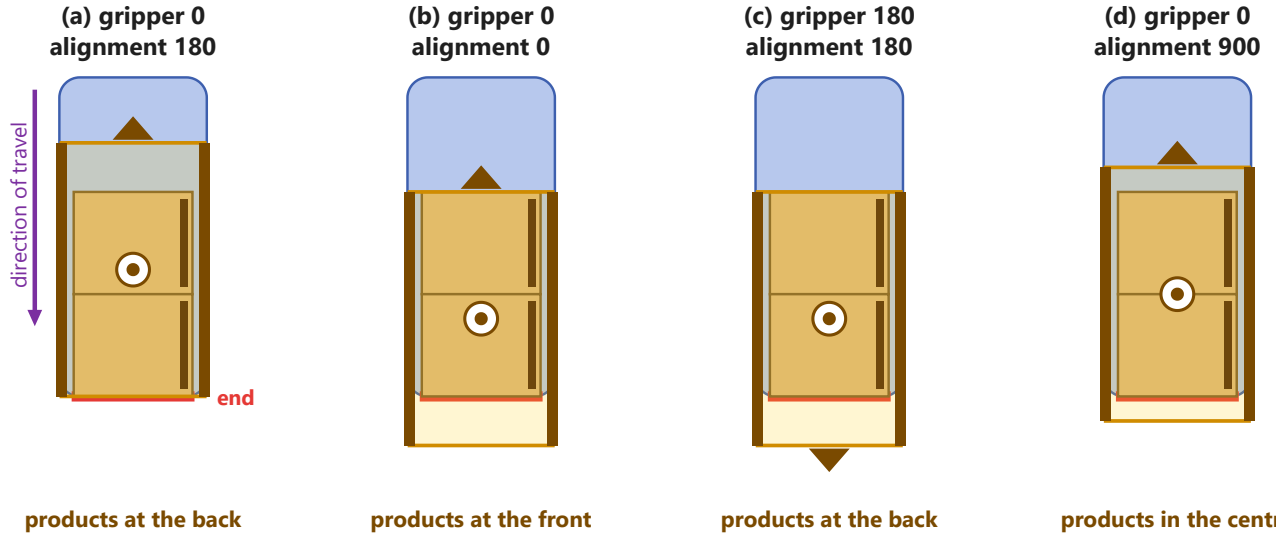


The open side (90 or 270) is the side of the gripper that stays free: with the opposite side the gripper sets the products against those already placed. The alignment (0, 180, 900) tells at which end of the gripper the products lie. Both belong to the gripper and count in its frame.



Gripper at orientation 0. Arrow: the open side.

- Open side (`gripper_open_side`): 90 or 270. With open side 270 the products lie against side 90, and with that side the gripper sets them against those already placed. Side 270 stays free, unless the station settings say otherwise.
- Jaw gripper: the side of the moving jaw. Fork gripper: the side the forks come in from. Another kind: the side with the parts that could collide with the neighbouring products.
- When both sides move, the station allows both (`gripper.allowed_open_side`).
- **900**: a gripper without an open side (from above). **999**: not assigned.
- Alignment (`gripper_align`): **0** = products at the gripper's front, **180** = at its back, **900** = in the centre.
- 900 is not a direction: in the alignment it means the gripper centred on the products.

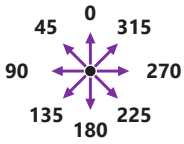


Alignment at the pick. The gripper's front is the triangle, its back the opposite end.

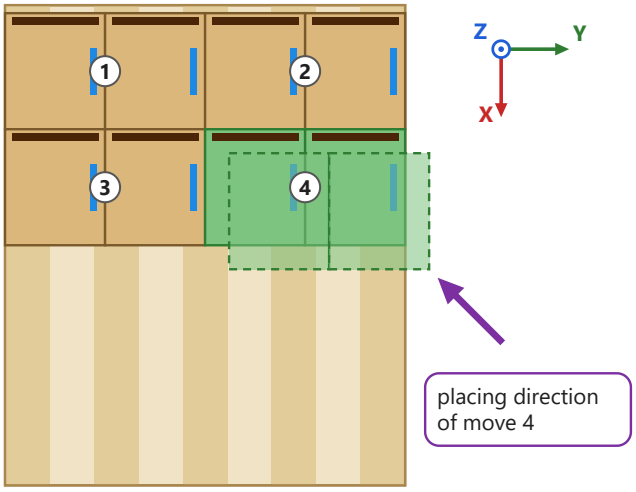
	JSON	PLC (`tPlacingData`, move data)	CSV column
open side	`gripper_open_side`: 90, 270, 900	`i16GripperOpenSide`: 90, 270, 900	`GripperOpenSide`
alignment	`gripper_align`: 0, 180, 900	`i16GripperAlign`: 0, 180, 900	`GripperAlign`

5 Placing direction

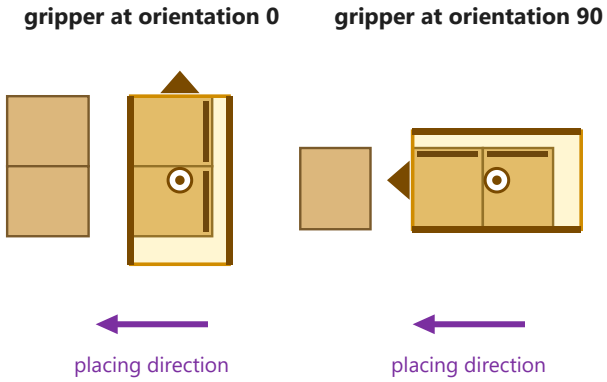
OpenPallet 1.2



The placing direction is the direction in which the gripper with the products travels in the last phase of its approach to the placing position, counted in the pallet frame. 900 means placing straight down from above.

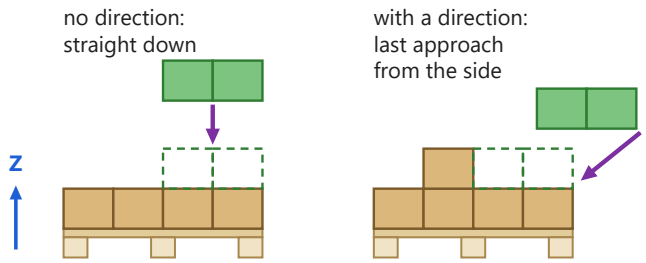


Product A, unique layer 1, move 4: placing direction 45, the products approach moves 2 and 3.



the same at both gripper orientations

Placing direction 90: the gripper sets the products to the left, against those already placed. It counts in the pallet frame, so it is the same at every gripper orientation.

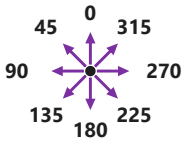


Side view.

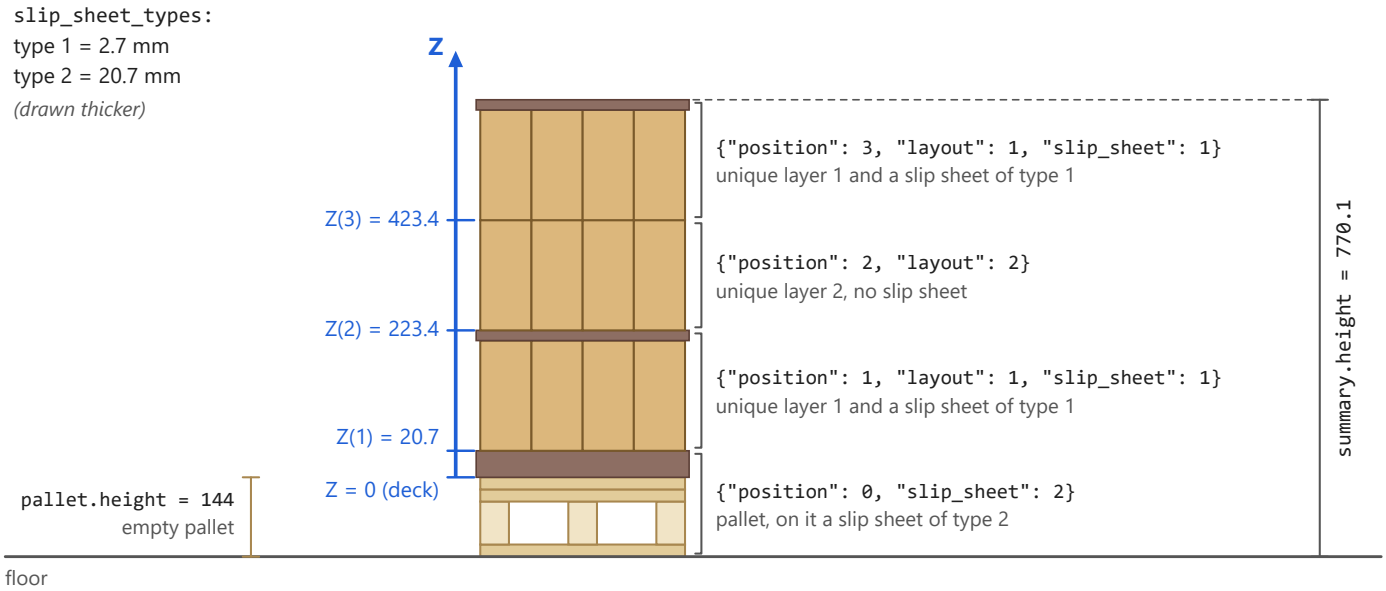
- **900**: straight down from above, without an approach from the side. **999**: not assigned.
- The robot program sets the height and the length of the approach: the file gives only the direction.

	JSON	PLC (`tPlacingData` , move data)	CSV column
placing direction	`placing_direction`: 0, 45, 90, … 315	`i16PlacingDirection`: 0, 45, … 315	`PlacingDirection`
straight down	900	900	900

6 Stack and slip sheets



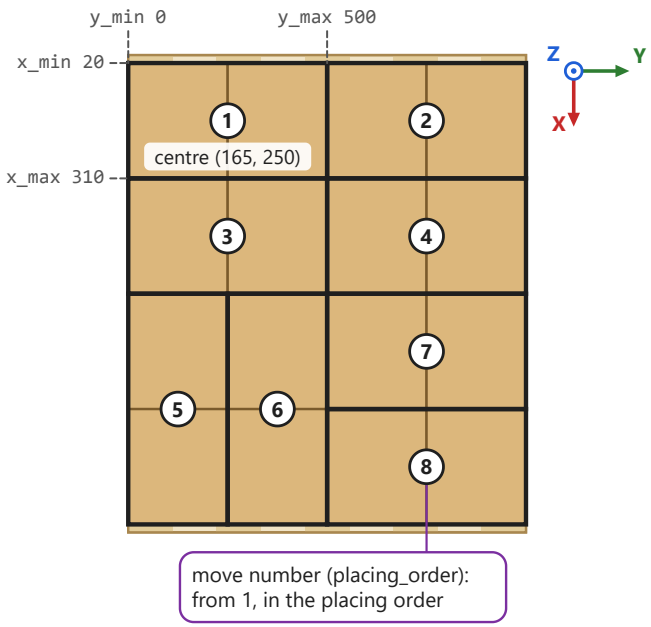
The stack (`stack`) lays the layers from the pallet up, each with a slip sheet or without one. Z counts from the pallet's deck boards, so the robot adds the height of the empty pallet (`pallet.height`) itself.



Example: two slip sheet types, three layers, product height 200 mm. Z in mm from the pallet's deck boards.

Z(K) = (K − 1) × product height + thicknesses of the slip sheets under layer K
summary.height = pallet.height + layers + all slip sheets

Another pallet height changes only `pallet.height` and `summary.height`. The Z of the layers stays, as it counts from the deck boards.

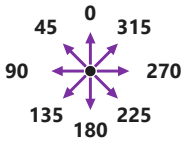


Product A, unique layer 1: the moves and the envelope of move 1 (the rectangle around its products).

- Slip sheet: a sheet on the pallet, between layers or on top.
- A stack position names a unique layer (`layout` = `id` from `layouts`) and the slip sheet lying on it (`slip_sheet` = `id` from `slip_sheet_types`). No `slip_sheet`: no slip sheet.
- A slip sheet type (`slip_sheet_types`, up to 10 types) has only a thickness (`thickness`): how much the slip sheet raises the next layer. The integrator sets its size and material.
- A unique layer (`layouts[]`): a layer pattern written once. The stack uses it in many layers of the pallet.
- A product of a move (`products[]`): `x`, `y` = the corner with the smaller X and Y, `size_x`, `size_y` = its size along X and Y.
- The centre of a move for the PLC (`tPlacingData`) is in whole numbers: trunc cuts off the fraction.

`i16MiddleX = trunc((x_min + x_max) / 2)` `i16MiddleY = trunc((y_min + y_max) / 2)`

7 OpenPallet 1.2 file



An OpenPallet file is JSON. Its top-level keys have a fixed order, and what belongs together lies nested in them: the moves in the unique layer, the products in the move.

{} **format**	name "OpenPallet", version "1.2"
{} **header**	recipe name, date saved, program, checksum
{} **pallet**	empty pallet: length (X), width (Y), height, in mm
{} **product**	product name and code
{} **summary**	summary: number of layers, products, moves, slip sheets, height, mass
[] **product_types**	product types
`id, name, code, length, width, height, mass, label_sides, count`	
[] **slip_sheet_types**	slip sheet types
`id, thickness`	
[] **stack**	stack: layers and slip sheets from the pallet (0) up
`position, layout → layouts.id, slip_sheet → slip_sheet_types.id`	
[] **layouts**	unique layers: every layer pattern written once
`id`	
[] **moves**	moves of the layer, in the placing order
`placing_order, x_min, x_max, y_min, y_max, x_middle, y_middle, size_x, size_y`	
`gripper_pick_orientation, gripper_place_orientation, placing_direction,`	
`gripper_align, gripper_open_side`	
[] **products**	products placed in this move: position on the pallet
`id, orientation, x, y, size_x, size_y, type → product_types.id`	
[] **products_at_pick**	the same products at the pick, from the conveyor end
`product_id, orientation`	
{} **gripper**	gripper of the station: outline, allowed orientations, open sides, alignments
{} **source**	conveyor: end (stop: 180), allowed product orientations
{} **limits**	controller limits: sizes of the PLC arrays
{} **x-palletdesigner**	PalletDesigner's data for opening the project in the program again. Other programs, the PLC and the robot may skip it.

Structure of the file in the order of writing. { }: object, [ ]: list, grey row: the fields of every item of the list.

- **Recognition:** `format.name` = `"OpenPallet"` and `format.version` 1.2 or a later 1.x: read. Another name or version: reject. Skip unknown keys.
- Later 1.x versions add only optional fields: a program that reads 1.2 reads every later 1.x file.
- Units: positions and sizes in mm, mass in g, orientations and directions in degrees (whole numbers). Slip sheet thickness and `summary.height` in mm with one decimal place.
- A pallet without a mix (`summary.mix` false): one product type, `id` 1, and the products in the moves have no field `type`.
- Checksum (`header.checksum`): CRC-32 (8 hex characters) of the JSON without it, keys sorted, no spaces, non-ASCII characters as `\uXXXX`, whole numbers without `.0`.